S-Parameter Analysis and Transmission Line Length Leo Marcinkus

Simulation Setup

S-parameters were extracted in Keysight ADS for a 50Ω-optimized microstrip transmission line. Both S11 (return loss) and S21 (insertion loss) were measured at 8 GHz for line lengths of 6, 12, 24, and 36 inches using frequency markers.

Summary Table - S11 and S21 at 8 GHz

Length	S11 at 8 GHz (Return Loss)	S21 at 8 GHz (Insertion Loss)	Observation
6 in	-43.79 dB	-3.46 dB	Excellent match, low insertion loss
12 in	-25.93 dB	-6.85 dB	Match degrades slightly, ~2x insertion loss
24 in	-24.63 dB	-13.67 dB	S21 doubles again with length
36 in	-24.56 dB	-20.49 dB	Heavy insertion loss at high frequency

S-Parameter Plots

The following plots show dB(S(1,1)) (return loss, blue) and dB(S(2,1)) (insertion loss, red) vs. frequency for each line length.

6-inch Transmission Line:

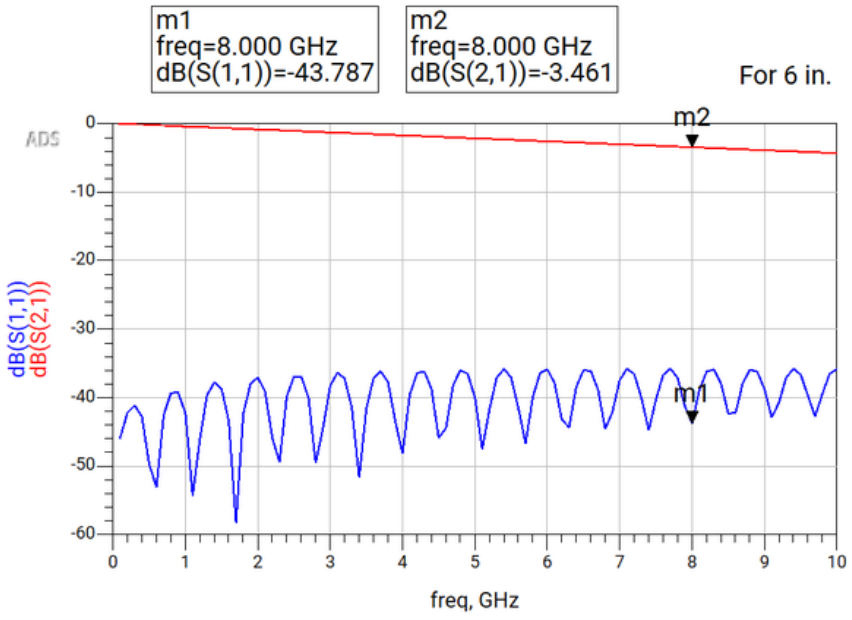


Figure 1: S11 = -43.79 dB, S21 = -3.46 dB at 8 GHz (6 in)

12-inch Transmission Line:

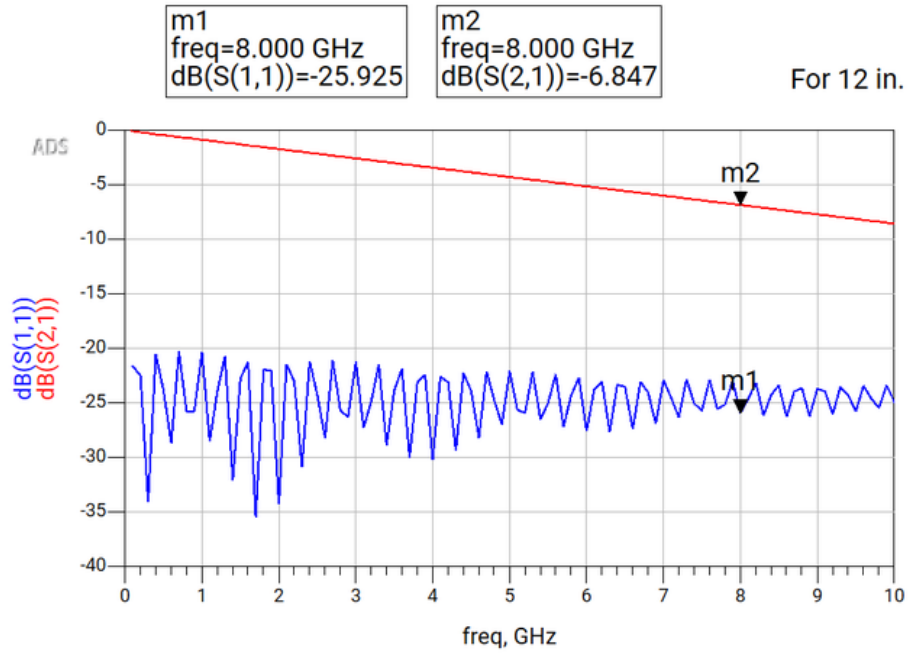


Figure 2: S11 = -25.93 dB, S21 = -6.85 dB at 8 GHz (12 in)

24-inch Transmission Line:

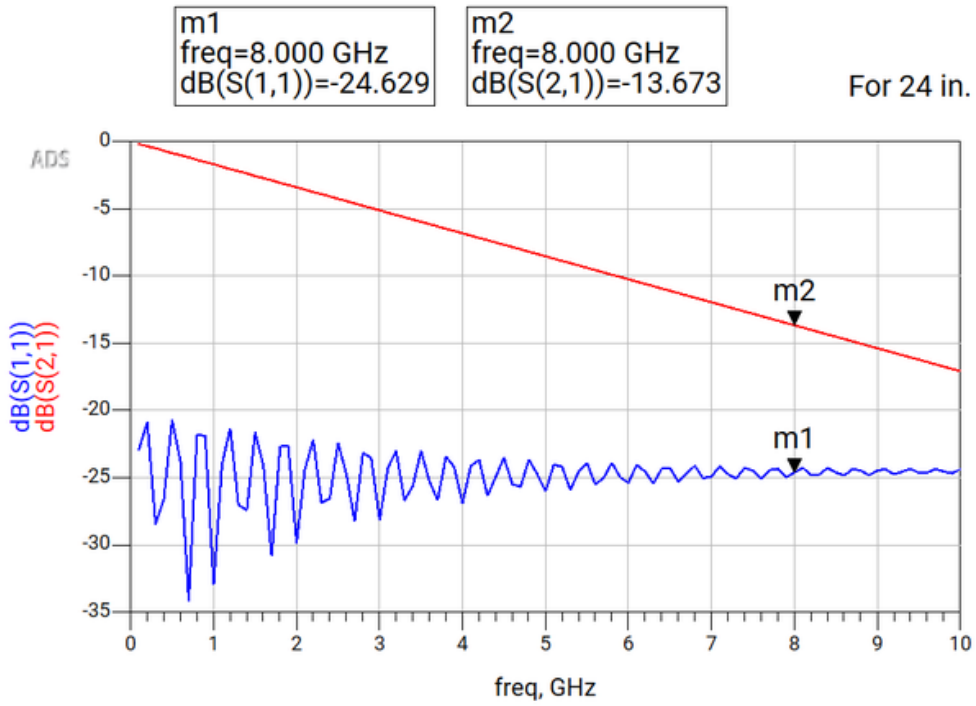


Figure 3: S11 = -24.63 dB, S21 = -13.67 dB at 8 GHz (24 in)

36-inch Transmission Line:

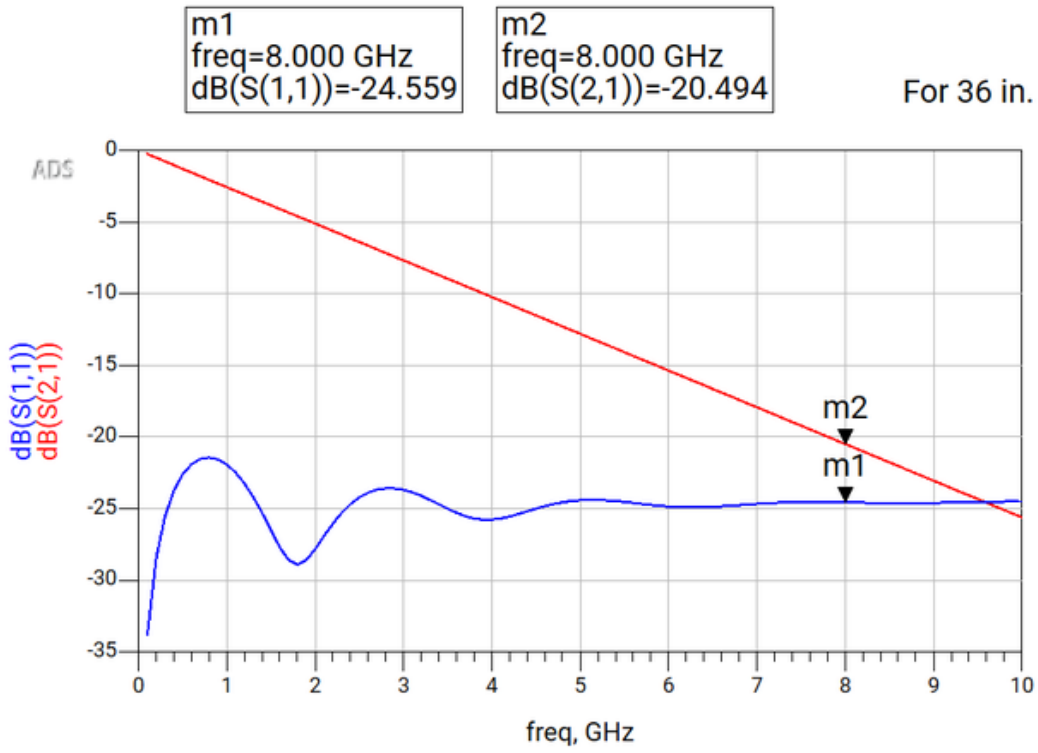


Figure 4: S11 = -24.56 dB, S21 = -20.49 dB at 8 GHz (36 in)